



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Sine, Cosine Rule & Area of Triangles

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- * Deciding the Trig Rule



Your notes

The Sine Rule

Sine Rule

What is the sine rule?

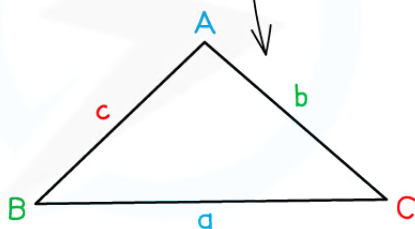
- The **sine rule** is used in **non right-angled triangles**
 - It allows us to find missing side lengths or angles
- It states that for any triangle with angles A , B and C

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

- Where
 - a is the side **opposite** angle A
 - b is the side **opposite** angle B
 - c is the side **opposite** angle C

LABEL YOUR TRIANGLE WITH CAPITALS FOR ANGLES AND LOWER CASE FOR THE OPPOSITE SIDE

ALWAYS RELABEL YOUR TRIANGLE ABC/abc TO MATCH THE RULE YOU NEED



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How do I use the sine rule to find missing lengths?

- Use the **sine rule**
 - when you have **opposite pairs** of sides and angles in the question



Your notes

- a and A , or b and B , or c and C
- **Start by labelling your triangle** with the angles and sides
 - Angles have upper case letters
 - Sides **opposite** the angles have the equivalent lower case letter
- To find a **missing length**, substitute numbers into the formula

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

- You only need to have two parts equal to each other (not all three)
 - Then solve to find the side you need

How do I use the sine rule to find missing angles?

- To find a **missing angle**, it is easier to **rearrange** the formula first by flipping each part

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

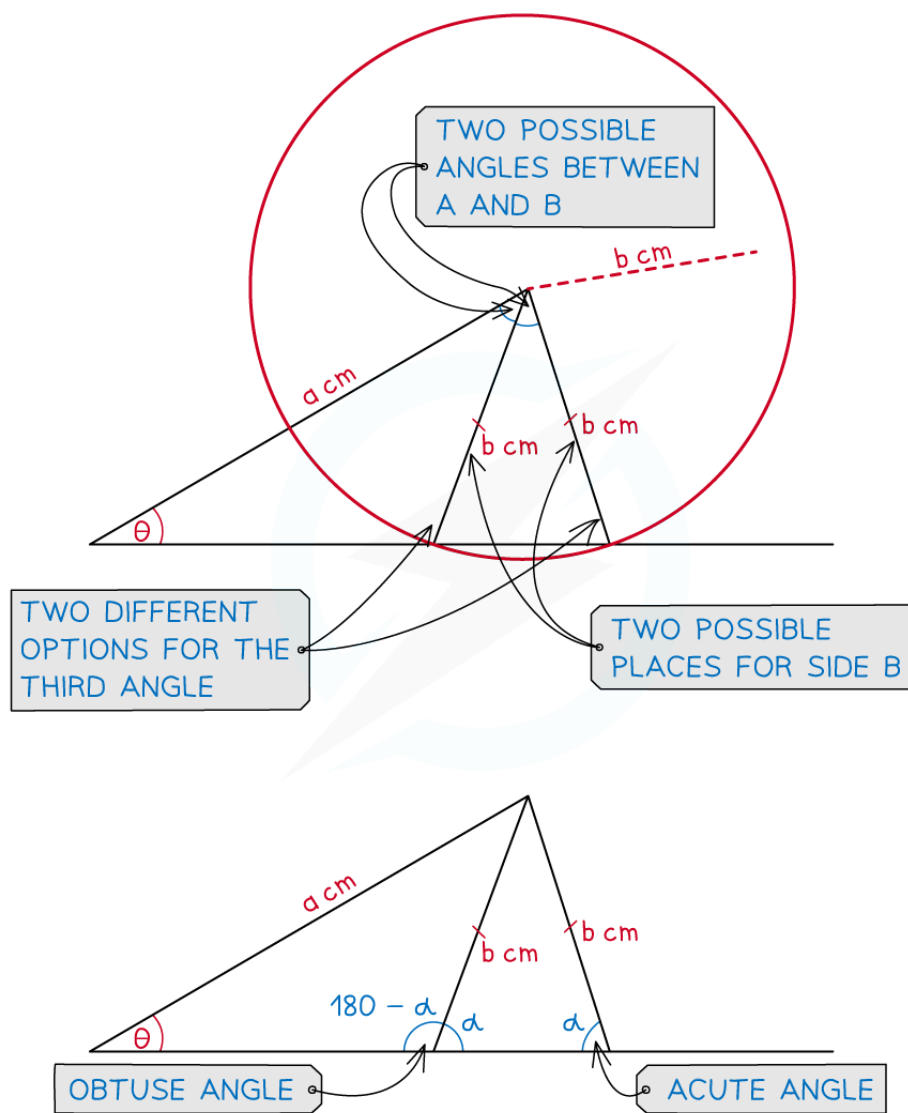
- The angles are now in the numerators of the fractions
- Substitute the values you have into the formula and solve
 - You will need to use **inverse sine** in your calculation, $\sin^{-1}(\dots)$

What is the ambiguous case of the sine rule?

- Given information about a triangle, there may be **two different ways** to draw it
- In the diagram below, the lengths of two sides are given, a and b
 - A base angle is also given, θ , but no angle near b is given
 - It turns out that there are **two possible ways** to arrange b to complete the triangle!
 - Both triangles have the correct values of a , b and θ
- To other base angle could either be **obtuse or acute**
 - The **sine rule** only gives the **acute** answer on your calculator
 - You need to check the diagram to see if the angle you need is actually obtuse
 - If it is, use this rule: **obtuse angle = $180 - \text{acute angle}$**



Your notes



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Examiner Tips and Tricks

- The sine rule is given in the formula booklet.



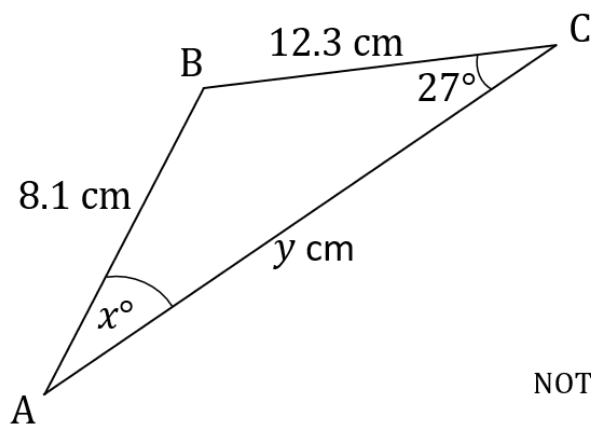


Your notes

Worked Example

The following diagram shows triangle ABC.

$AB = 8.1 \text{ cm}$, $BC = 12.3 \text{ cm}$ and angle $BCA = 27^\circ$.



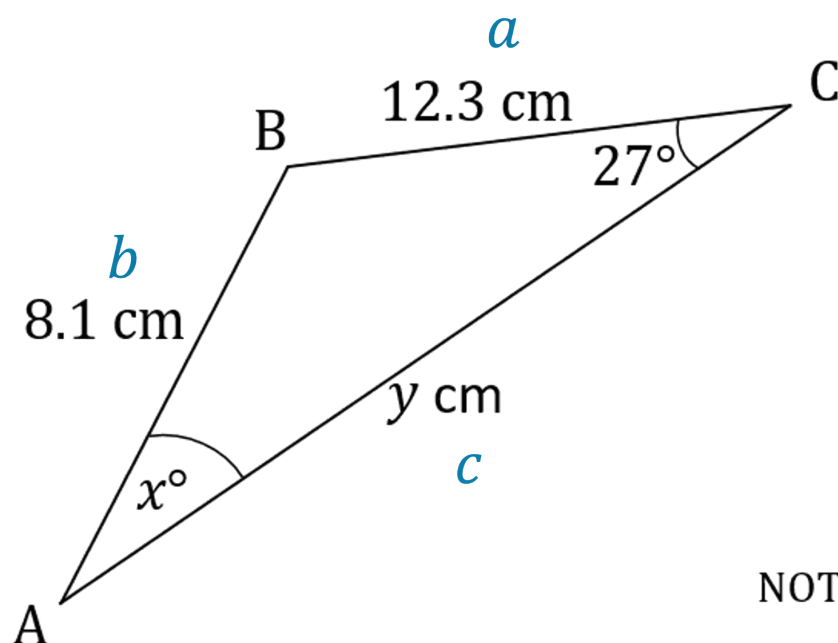
NOT TO SCALE

(a) Calculate the value of x .

Label the sides of the diagram



Your notes



NOT TO SCALE

x is an angle so use the sine rule with the angles on top

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

In practice, you only need to equate two of these three parts

$$\frac{\sin x}{12.3} = \frac{\sin 27}{8.1}$$

$$\sin x = \frac{12.3 \sin 27}{8.1}$$

$$x = \sin^{-1}\left(\frac{12.3 \sin 27}{8.1}\right)$$

$$x = 43.58207...$$

$$43.6^\circ \text{ (to 1 d.p.)}$$

(b) Calculate the value of y.

To find y you need to know the angle opposite (angle ABC)

You know 27 and x from above, so subtract these from 180



Your notes

$$\begin{aligned}\text{Angle } ABC &= 180 - 27 - 43.58207... \\ &= 109.41792...\end{aligned}$$

y is a length so use the sin rule with the sides on the top

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{y}{\sin(109.41792...)} = \frac{8.1}{\sin 27}$$

$$y = \frac{8.1 \sin(109.41792...)}{\sin 27}$$

$$y = 16.82691...$$

$$\mathbf{y = 16.8 \text{ cm (to 3 s.f.)}}$$



Your notes

The Cosine Rule

Cosine Rule

What is the cosine rule?

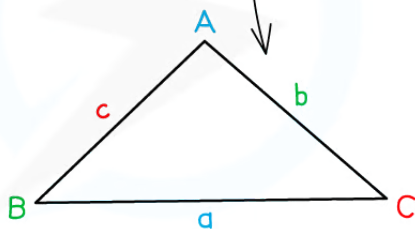
- The **cosine rule** is used in **non right-angled triangles**
 - It allows us to find missing side lengths or angles
- It states that for any triangle

$$a^2 = b^2 + c^2 - 2bc \cos A$$

- Where
 - **a** is the side **opposite** angle **A**
 - **b** and **c** are the other two sides
 - **b** and **c** are either side of angle **A**
 - **A** is the angle between them

LABEL YOUR TRIANGLE WITH CAPITALS FOR ANGLES AND LOWER CASE FOR THE OPPOSITE SIDE

ALWAYS RELABEL YOUR TRIANGLE ABC/abc TO MATCH THE RULE YOU NEED



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How do I use the cosine rule to find a missing length?

- Use the **cosine rule** for lengths
 - when you have **two sides and the angle between them**



Your notes

- and you want to find the **opposite side**, a
- **Start by labelling your triangle** with the angles and sides
 - Angles have upper case letters
 - Sides **opposite** the angles have the equivalent lower case letter
- **Substitute** values into $a^2 = b^2 + c^2 - 2bc \cos A$
 - Make a the subject (don't forget to **square root**)

How do I use the cosine rule to find a missing angle?

- Use the **cosine rule** for angles
 - when you have **all three sides**
 - and you want to find an angle
- It helps to **rearrange** the formula as follows, by adding $2bc \cos A$ to both sides then making $\cos A$ the subject

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$a^2 + 2bc \cos A = b^2 + c^2$$

$$2bc \cos A = b^2 + c^2 - a^2$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

- Use the formula $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ to find the **unknown angle** A
 - Remember, A is the angle **between** sides b and c
 - (you may need to relabel the triangle)
 - You will need to use **inverse cosine** at the end, $\cos^{-1}(\dots)$
- Unlike the sine rule, there is no ambiguous case of the cosine rule



Examiner Tips and Tricks

- You are given the cosine rule in the form $a^2 = b^2 + c^2 - 2bc \cos A$ on the formula sheet

- Getting an error on your calculator when finding an angle may mean you have rearranged the formula incorrectly

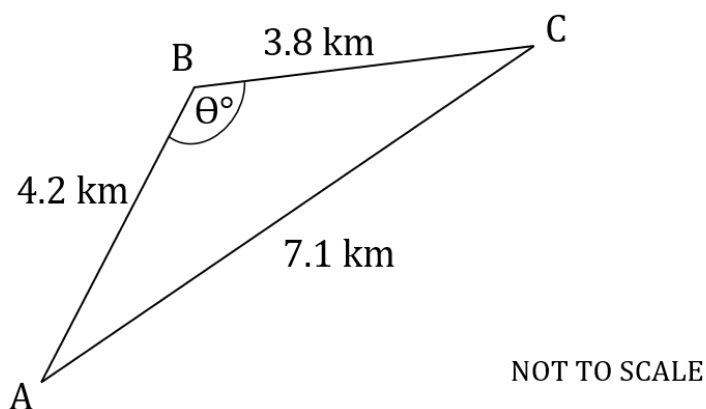


Your notes



Worked Example

The following diagram shows triangle ABC , where $AB = 4.2$ km, $BC = 3.8$ km and $AC = 7.1$ km.



Calculate the value of angle ABC .

The side opposite the angle is 7.1, so $a = 7.1$

The sides 4.2 and 3.8 are b and c (in either order)

Use the cosine rule in the rearranged form $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

$$\begin{aligned}\cos \theta &= \frac{4.2^2 + 3.8^2 - 7.1^2}{2(4.2)(3.8)} \\ \theta &= \cos^{-1}\left(\frac{4.2^2 + 3.8^2 - 7.1^2}{2(4.2)(3.8)}\right) \\ \theta &= 125.04699\dots\end{aligned}$$

$$\theta = 125.0^\circ \text{ (to 1 d.p.)}$$



Your notes



Your notes

Area of a Triangle

Area of a Triangle

How do I find the area of a non right-angled triangle?

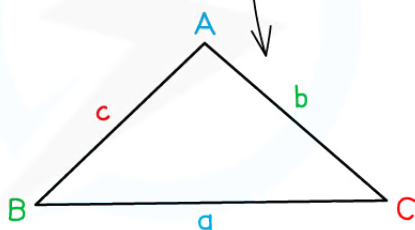
- The area of **any triangle** can be found using the formula

$$\text{Area} = \frac{1}{2} ab \sin C$$

- C is the angle **between** sides *a* and *b*

LABEL YOUR TRIANGLE
WITH CAPITALS FOR
ANGLES AND LOWER
CASE FOR THE
OPPOSITE SIDE

ALWAYS RELABEL YOUR
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- Label your triangle correctly
 - Make sure that C is always the angle **between** the two sides
- If angle C is 90° , you get a right-angled triangle
 - $\sin 90^\circ = 1$ so the formula becomes the familiar "Area = $\frac{1}{2}$ × base × height"!



Examiner Tips and Tricks

- You are given the triangle area rule on the formula sheet



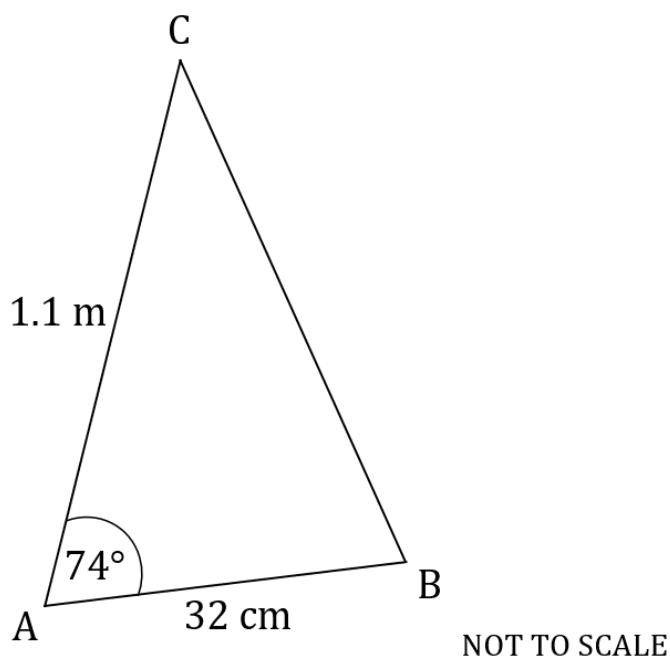
Your notes



Worked Example

The following diagram shows triangle ABC .

$AB = 32 \text{ cm}$, $AC = 1.1 \text{ m}$ and angle $BAC = 74^\circ$.



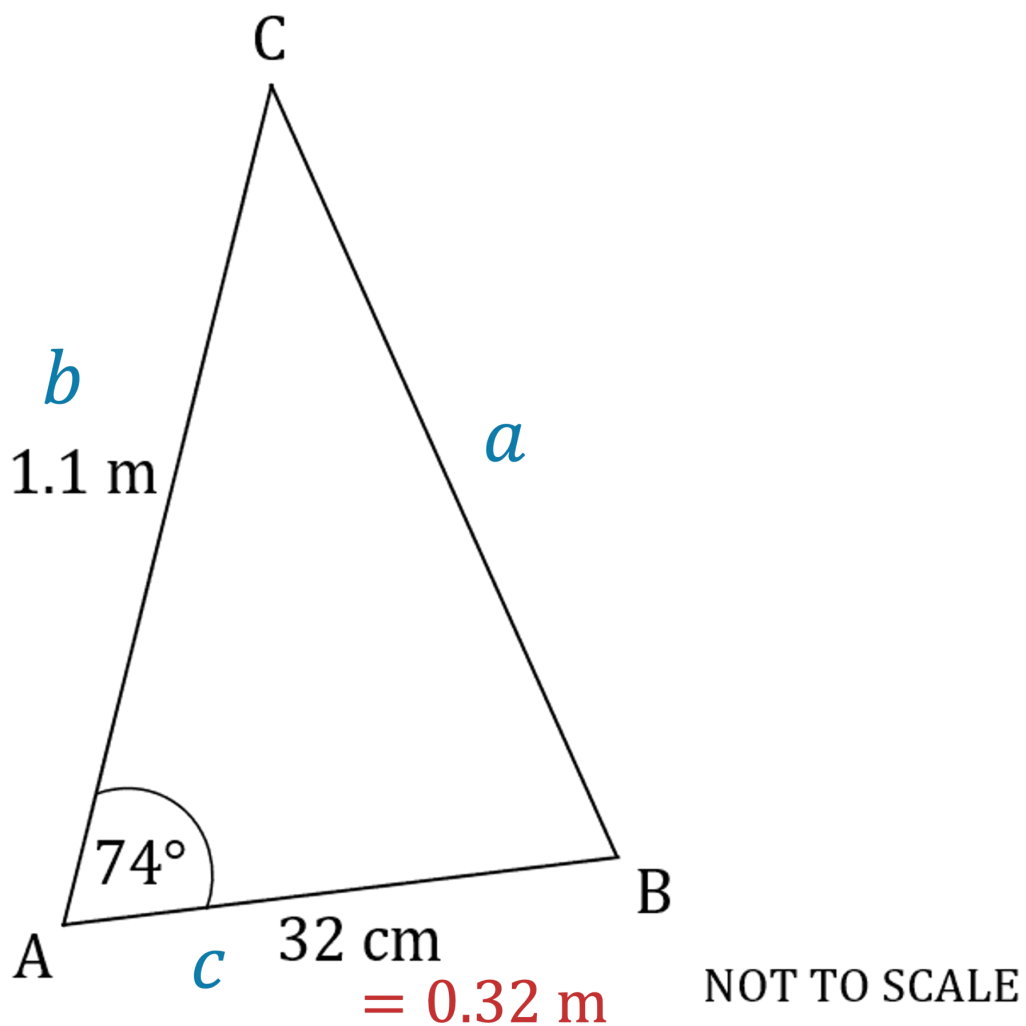
Calculate the area of triangle, giving your answer in m^2 .

Label the sides of the triangle

Convert the sides to be in the same units



Your notes



Use the area of a triangle formula, $\text{Area} = \frac{1}{2}bc \sin A$

$$\begin{aligned}\text{Area} &= \frac{1}{2}(1.1)(0.32) \sin 74 \\ &= 0.16918...\end{aligned}$$

The area is 0.169 m^2 (to 3 s.f.)



Your notes

Deciding the Trig Rule

Applications of Trigonometry

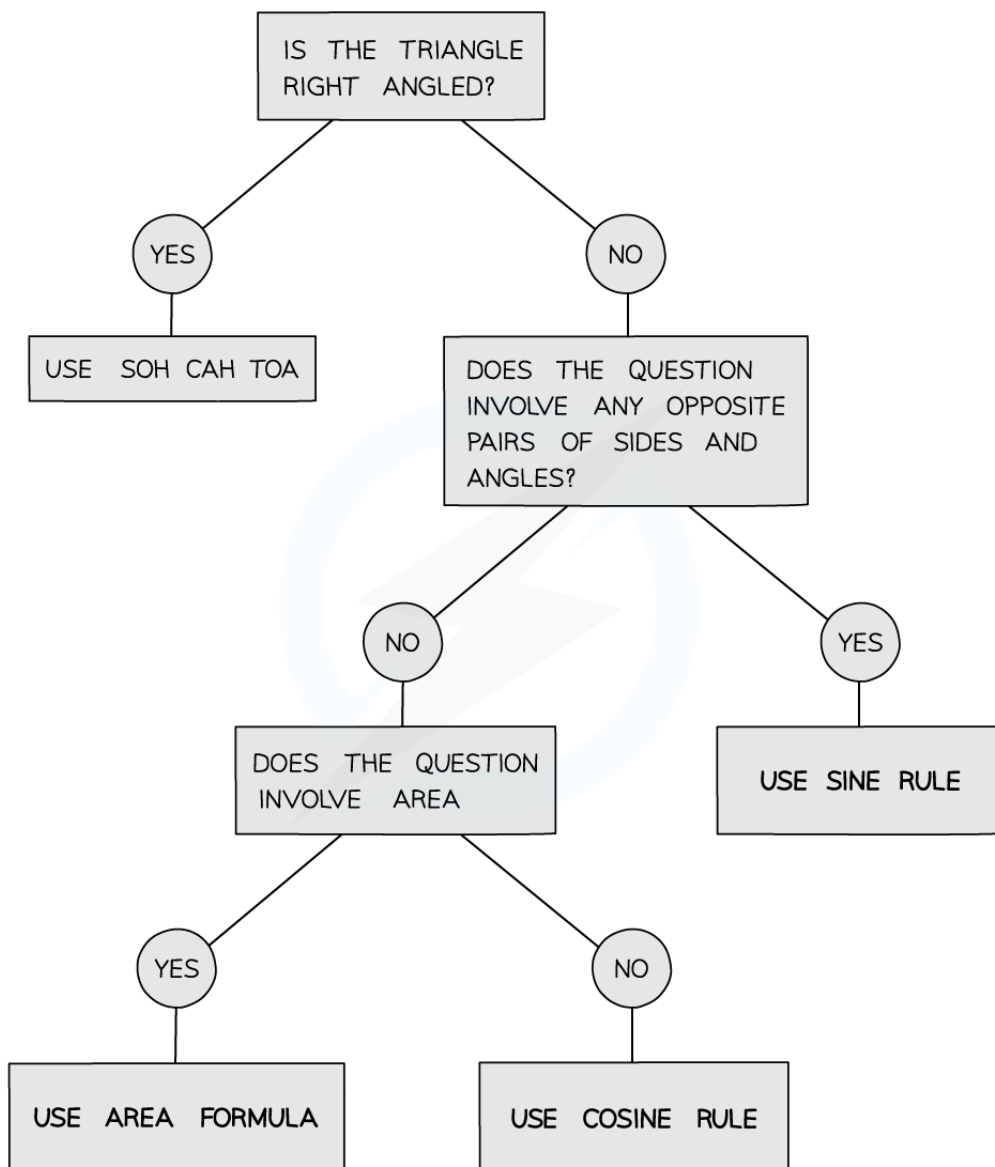
How do I decide which trig rule to use?

- **Different rules** are required depending on the question
 - You need to be able to decide which is appropriate to use
 - Think about what **information** you **have** and what you want to **find**
- This table summarises the possibilities:

If you know	And you want to know	Use
Two sides and an angle opposite one of the sides	The angle opposite the other side	Sine rule
Two angles and a side opposite one of the angles	The side opposite the other angle	Sine rule
Two sides and the angle between them	The third side	Cosine rule
All three sides	Any angle	Cosine rule
Two sides and the angle between them	The area of the triangle	Area of a triangle rule



Your notes



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Can I use multiple trig rules in the same question?

- Harder questions will require you to use **more than one trig rule**
 - For example, you may need the sine rule **followed by** the cosine rule
- The **area formula** only works for an angle between two sides

- If you are **not** given this setup, you may need to use the sine or cosine rule first
- If it looks like no rule would work, remember that all **angles in a triangle sum to 180**
 - This often helps to find a missing angle



Your notes



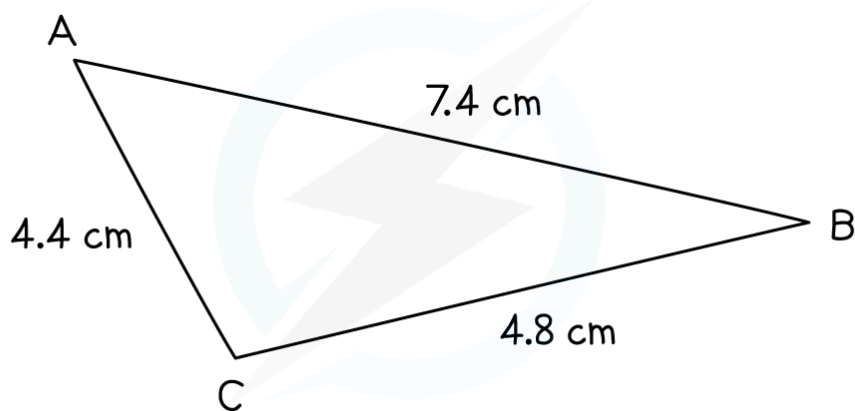
Examiner Tips and Tricks

- Look at the number of marks for a question – if it is a lot, you are likely to need more than one trig rule!



Worked Example

Find the area of the triangle below.



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The area of a triangle can be found using the formula $A = \frac{1}{2} ab \sin C$

The three side lengths are known, but we need to find an angle in order to calculate the area
Because we know all three sides, any of the angles could be found

Find angle ABC using the cosine rule



Your notes

Cosine Rule: $a^2 = b^2 + c^2 - 2bc \cos A$,

where A is the angle opposite side a

$$AC^2 = AB^2 + BC^2 - 2(AB)(BC) \cos ABC$$

$$4.4^2 = 7.4^2 + 4.8^2 - 2(7.4)(4.8) \cos ABC$$

Rearrange to make $\cos ABC$ the subject

$$4.4^2 - (7.4^2 + 4.8^2) = -2(7.4)(4.8) \cos ABC$$

$$\cos ABC = \frac{4.4^2 - 7.4^2 - 4.8^2}{-2(7.4)(4.8)}$$

Use your calculator to find the value of $\cos ABC$

$$\cos ABC = \frac{487}{592}$$

Use the $\cos^{-1}$ button on your calculator to find the value of ABC

$$\begin{aligned} ABC &= \cos^{-1}\left(\frac{487}{592}\right) \\ &= 34.65054... \end{aligned}$$

Now we can find the area of the triangle using the formula and angle ABC as the known angle

$$\begin{aligned} \text{Area} &= \frac{1}{2} \times 4.8 \times 7.4 \times \sin 34.65054... \\ &= 10.09779... \end{aligned}$$

$$\text{Area} = 10.1 \text{ cm}^2 \text{ (3 s.f.)}$$